MIL-I-81496(AS)
AMENDMENT 1
15 October 1968

MILITARY SPECIFICATION

INDICATOR, ATTITUDE, 2-INCH

ID-1602/A

This amendment forms a part of Military Specification MIL-I-81496(AS) dated 15 April 1968, and has been approved by the Naval Air Systems Command, Department of the Navy.

Page 2, Under Military: Add

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|---|---|
| "MIL-M-7793 | Meter, time totalizing |
| MIL-C-14806 | Coating, Reflection reducing for instrument cover glasses and lighting wedges |
| MIL-C-26482 | Connectors, Electric, Circular, Miniature quick disconnect, environment testing |
| MIL-G-81559 | Gyroscope Assemblies, Attitude and Directional Reference Instructions for Aircraft: Preservation and Packaging, procedures for" |

Page 3, under Military: Delete "MIL-STD-794" and title and add:

| | |
|---|---|
| "MIL-STD-838 | Lubrication of Military Equipment" |

Page 5, Paragraph 3.3: Add

"The sphere shall not oscillate or jitter more than ± 1/4 degree when subjected to 2G vibration".

Page 6: Add

"3.3.2.4 Time Totalizing Meter - The indicator shall contain a time totalizing meter (4,000 hours) conforming to Specification MIL-M-7793. The reading of the meter shall not require unsealing of the indicator case. The time totalizer shall be installed within the present envelope of the instrument."

[FSC 6610]

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Page 7, paragraph 3.3.9.1: Delete and substitute:

"3.3.9.1 Vibration - The indicator shall operate satisfactorily without loose parts or internal damage when subjected to the vibration tests of 4.8.21 and 4.8.22."

Page 8, paragraph 3.4.1.1: Delete sentence and add:

"3.4.1.1.1 Standard Conditions - The indicator, when connected and operated at standard conditions, shall provide a pitch and roll accuracy of 0.25 degree minimum.

3.4.1.1.2 Longitudinal Acceleration - The indicator, when subjected to an 8G acceleration for 4 seconds along the roll axis, shall provide a pitch and roll accuracy of 2 degrees minimum.

3.4.1.1.3 Coordinated Turns - The indicator, when subjected to coordinated turns with rates of turn between 90 and 1500 degrees per minute and bank angles of 20 to 60 degrees, shall provide a pitch and roll accuracy of 3 degrees minimum.

3.4.1.1.4 Uncoordinated Turns - The indicator, when subjected to uncoordinated turns (skid turns) of less than 5 degrees per minute at air speeds not greater than 400 knots for a 10-minute duration, shall provide a pitch and roll accuracy of 2 degrees minimum."

Page 14, paragraph 3.5.5.1 and Table: Delete and substitute:

"3.5.5.1 Antireflection Coating - All reflecting glass surfaces of the cover glass and lighting wedge shall be covered with a reflection reduced coating that meets the requirements of MIL-C-14806 in addition to withstanding the environmental requirements specified herein. The humidity requirements shall be goverened by paragraph 4.8.31."

Page 14, paragraph 3.5.6.2 line 5: Delete "White, color No. 37875" and substitute:

"Fluorescent Red, Color No. 11105 of FED-STD-595."

Page 15, add:

"3.7 Lubrication - The gyro bearings shall be lubricated with MINIPURE grease or equivalent with particle size not exceeding 25 microns for 2% and 5 microns for 98% of volume. The requirements of MIL-STD-838 shall be met."

Page 26, paragraph 4.8.6: Add:

"The test shall be repeated with power removed."



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Page 30, paragraph 4.8.22: Delete and substitute:

"4.8.22 Vibration Failure - The indicator shall be tested in accordance with Part II Vibration Tests of MIL-T-5422 with power applied except Curve II of Fig. 2 shall be extended at 2g to 20 cycles per second and at 350 cycles per second it shall be changed to 5g and extended to 2000 cycles per second. At the conclusion of the test the equipment shall meet the requirements of the individual tests."

Page 30, paragraph 4.8.24.1: Delete and substitute:

"4.8.24.1 Operating - Rated power shall be applied to the equipment and precessions in pitch and roll resulting from accelerations shall be limited as stated in 3.4.1.1."

Page 30, Add:

"4.8.24.3 Longitudinal Accelerations - The equipment shall meet the requirements of 3.4.1.1.2.

4.8.24.4 Coordinated Turns - The equipment shall meet the requirements of 3.4.1.1.3.

4.8.24.5 Uncoordinated Turns - The equipment shall meet the requirements of 3.4.1.1.4"

Page 30, paragraph 4.8.25, third and fourth sentences: Delete and substitute:

"The error in either pitch and roll shall not exceed 3°. The error induced by roll maneuvers at a rate of 300° ±10° per second shall not exceed one degree. Furthermore, rotating the indicator in either direction at the above rates shall not cause damage to the indicator."

Page 32: Add:

"4.8.34 Sand and Dust - The indicator shall be subjected to sand and dust tests procedure I of Specification MIL-E-5272. At the end of the sand exposure, the indicator shall not be damaged and shall meet the requirements of the individual tests."

Pages 33 and 34, Paragraphs 5.1, 5.2, 5.3, 5.4, and 5.4.1:



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Delete and substitute:

"5.1 PACKAGING, PACKING, PRESERVATION and MARKING - Shall be in accordance with MIL-G-81559."



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